GridShift: Electricity Demand Forecasting

Phases 1-4 | Dataset, model design, results, and contributions

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Executive summary. This report documents the forecasting task and data setup (Phases 1-2), model architecture and experimental procedure (Phase 3), and observed results (Phase 4). On the same 25,678 complete hourly rows and chronological 70/15/15 split, the selected TFT achieved validation MAE 1,874.62 MW and test MAE 3,542.41 MW. The selected LSTM achieved validation MAE 2,128.01 MW and test MAE 4,258.34 MW. The supplied LSTM run was CPU-based and selected the small configuration by validation MAE.

Phase 1: Problem and research scope

The task is one-hour-ahead forecasting of hourly electricity demand (MW) from historical demand, solar generation, wind generation, and calendar features. The intended users are grid-planning analysts, renewable-energy researchers, and forecasting practitioners. This is an offline research comparison, not a dispatch or control system. Forecast error is reported with MAE and RMSE; classification accuracy is not applicable to this regression task.

Phase 2: Data and split

The source is the hourly India electricity demand, solar, and wind dataset by Mukherjee, Kalita, and Kumar (Mendeley Data, Version 2; DOI: 10.17632/y58jknpgs8.2). The modeling CSV is `grid_hourly_merged.csv`. Complete rows were retained and ordered chronologically. TFT and LSTM each used 25,678 rows, with 17,974 training, 3,852 validation, and 3,852 test rows. Model scaling was fitted on training data only. The validation period selected configurations; the test set was held out for final reporting.

Phase 3: Model design and training

TFT used a 168-hour encoder and a larger configuration selected by validation MAE; training used GPU. LSTM used a 168-hour lookback with nine input features (demand, solar, wind, and six cyclic calendar encodings). Its three tested configurations were small, balanced, and deeper. The small model (32 hidden units, one layer, no dropout, learning rate 0.001, batch size 128) was selected by validation MAE. The supplied run trained for 30 epochs on CPU; its best validation MAE occurred at epoch 27. Standardization parameters were estimated from training rows only.

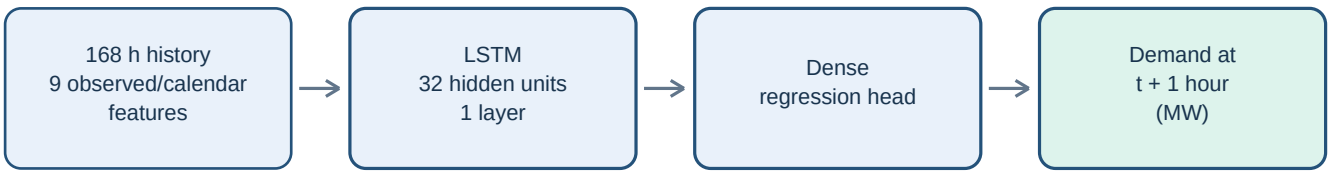
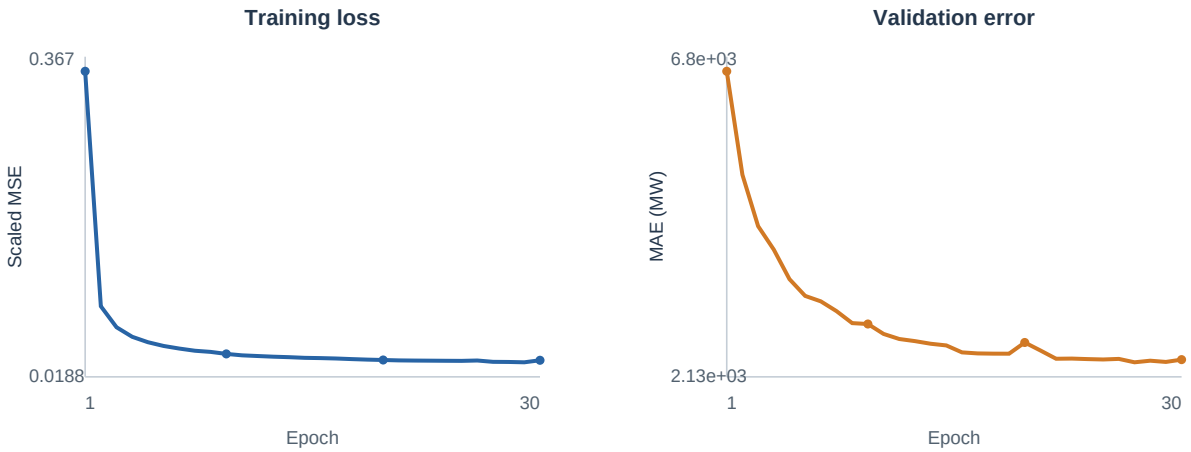


Figure 1. Selected LSTM information flow. Each forecast uses the prior 168 hours and predicts demand at the next hour.

Training history

The user-provided history contains scaled training MSE and validation MAE. They have different units and are shown in separate panels. For regression, validation MAE/RMSE curves replace classification accuracy curves. Training MAE/RMSE were not recorded in the supplied run.



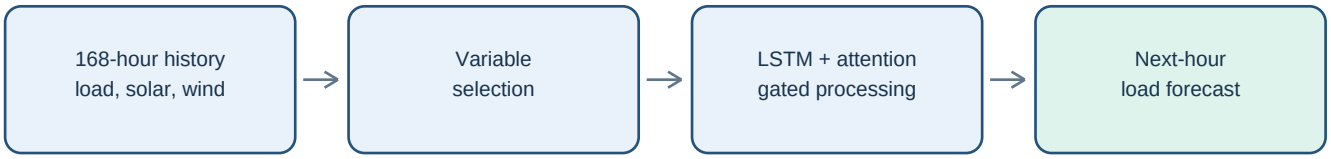
Separate axes: training loss is standardized MSE; validation metric is MW MAE.

TFT learning curve (selected trial)



Figure 2. Training and validation losses logged by the TFT run. See the source plot for axis labels and scale.

TFT forecasting architecture



Selected larger TFT configuration; GPU training

Figure 3. High-level sequence encoding, variable selection, attention, and forecast path.

Phase 4: Results

The reported figures below use the same target, one-hour horizon, row count, and chronological split for TFT and LSTM. Values are in MW. TFT was trained on GPU; LSTM was trained on CPU. Model selection used validation MAE; test metrics are final held-out results.

Model / split	MAE (MW)	RMSE (MW)	Configuration / note
TFT - validation	1,874.62	3,616.16	Larger; selected by validation MAE
TFT - test	3,542.41	6,610.82	Held-out period
LSTM - validation	2,128.01	3,770.96	Small; selected by validation MAE
LSTM - test	4,258.34	6,929.33	Held-out period
LSTM - training	Not recorded	Not recorded	Only scaled training MSE was logged by the supplied run.

On this split, TFT has lower reported validation and test MAE/RMSE than the supplied LSTM run. The test MAE is higher than validation MAE for both models, so generalization to the later period is weaker. These results describe this one chronological holdout and do not by themselves establish a causal effect from solar or wind generation.

Contribution table

The table maps project modules to their implementation files.

Project module	Implementation files
Data audit and cleaning	audit_india_dataset.py; clean_india_grid.py
TFT model	src/models/tft_model.py
XGBoost implementation (excluded from current findings)	src/models/xgboost_model.py
LSTM model	src/models/lstm_model.py
Report and documentation	gridshift_phase1_phase4.tex; README.md

Limitations

The supplied LSTM run did not report training MAE or RMSE; the available history contains scaled training MSE and validation MAE only. Results reflect one chronological holdout. The earlier XGBoost run used 12,984 rows and is excluded from the 25,678-row TFT/LSTM findings.

References

Mukherjee, D., Kalita, K., & Kumar, S. (2025). *Electricity Demand, Solar and Wind Generation Data (September 2021-June 2025) of India at 1-hour interval* (Version 2). Mendeley Data. DOI: 10.17632/y58jknpgs8.2. Hochreiter, S. & Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation, 9*(8), 1735-1780. Lim, B. et al. (2021). Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting, 37*(4), 1748-1764.